



FACULTY OF COMPUTING

SEMESTER 2, SESSION 2025/2026

SECI1143 – PROBABILITY & STATISTICAL DATA ANALYSIS

ASSIGNMENT 3

SECTION 02

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Question 1:

$$H_0 = p_1 = p_2$$

$$\text{Critical value} = 1.96$$

$$H_1 = p_1 \neq p_2$$

$$\hat{p}_v = 105/150 = 0.7$$

$$Z = \frac{0.7 - 0.55}{\sqrt{\frac{0.6333(1-0.6333)}{150 + 120}}}$$

$$\hat{p}_{nv} = 66/120 = 0.55$$

$$\hat{p}_{\text{both}} = \frac{105 + 66}{150 + 120} = 0.6333$$

$$Z = 2.542$$

$\therefore$ Since $2.542 > 1.96$, we reject H_0 .

It falls in the critical region.

There is sufficient evidence to support the claim

that there is significant difference between the proportions

of vaccinated individuals who agreed to participate in the

free health screening programme

Question 2

a) Let old machine $n_1 = 12$, $s^2_1 = 20$
Let new machine $n_2 = 14$, $s^2_2 = 9$

$$H_0: \sigma^2_1 = \sigma^2_2$$

$$H_1: \sigma^2_1 > \sigma^2_2$$

b)

$$F = \frac{s^2_1}{s^2_2}$$
$$= 20 \div 9$$
$$= 2.222 \text{ (3 dp)}$$

c)

$$\begin{array}{ll} \text{Numerator } df_1 = 12 - 1 & \text{Denominator } df_2 = 14 - 1 \\ = 11 & = 13 \end{array}$$

$$F_{\text{crit}} = F_{\alpha, df_1, df_2}$$
$$= F_{0.05, 11, 13}$$
$$\approx 2.6346$$
$$= 2.635 \text{ (3 dp)}$$

d)

Since the calculated test statistic ($F = 2.222$)
< critical value ($F_{\text{crit}} = 2.635$), we fail to
reject the null hypothesis, H_0

There is insufficient evidence at $\alpha = 0.05$ significance
level to conclude that the variance of the new machine is
significantly smaller than the variance of the old machine

Question 3

a) Let Programme A: $s_1 = 1.8 \Rightarrow s_1^2 = 3.24$
Programme B: $s_2 = 1.5 \Rightarrow s_2^2 = 2.25$

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 > \mu_2$$

b)

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$
$$= \frac{(14 - 1)(1.8)^2 + (12 - 1)(1.5)^2}{14 + 12 - 2}$$
$$= \frac{42.12 + 24.75}{24}$$
$$= 2.7863$$
$$= 2.786 \text{ (3 dp)}$$

$$t_0 = \frac{\bar{x}_1 - \bar{x}_2 - 0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$
$$= \frac{6.5 - 5.2}{2.7863 \sqrt{\frac{1}{14} + \frac{1}{12}}}$$
$$= \frac{1.3}{0.6567}$$
$$= 1.980 \text{ (3 dp)}$$

$$\begin{aligned} c) \quad df &= n_1 + n_2 - 2 \\ &= 14 + 12 - 2 \\ &= 24 \end{aligned}$$

$$\begin{aligned} t_{\alpha, df} &= t_{0.05, 24} \\ &= 1.711 \quad (3dp) \end{aligned}$$

d) Since our calculated test statistic ($t = 1.980$) > critical value ($t_{0.05, 24} = 1.711$), we reject the null hypothesis, H_0

There is sufficient evidence at the $\alpha = 0.05$ significance level to conclude that Diet Programme A produces a significantly greater mean weight loss than Diet Programme e

Q4

a) Hypothesis

H_0 : Number of borrowed books is uniformly distributed across weekdays

H_1 : Number of borrowed books is not uniformly distributed across weekdays

b) Expected Frequency & Test Statistic (5 marks)

$$E_i = \frac{\text{total}}{k} = \frac{210}{5} = 42$$

Day	O	E	O-E	(O-E) ²	(O-E) ² /E
Mon	42	42	0	0	0.000
Tue	38	42	-4	16	0.381
Wed	45	42	3	9	0.214
Thu	36	42	-6	36	0.857
Fri	50	42	8	64	1.524
Total	210	210			3.286

$$\chi^2 = \sum \frac{(O-E)^2}{E} = 3.286$$

c) Critical Value

Degrees of freedom:

$$df = k - 1 = 5 - 1 = 4$$

From the Chi-Square table at $\alpha = 0.05$, $df = 4$

$$\chi^2_{0.05, 4} = 9.488$$

d) Conclusion

Reject H_0 if χ^2 (calculated) $>$ χ^2 (critical)

compare: 3.286 < 9.488

Decision : Fail to reject H_0

Conclusion : At $\alpha = 0.05$, there is insufficient evidence to conclude that the number of books borrowed differs from weekdays.

a)

H_0 : variables are dependent

H_1 : variables are related.

b)

Online shopping frequency	Age 18-35	Exp	Age 36+	Exp	Total
Rarely	25	42.55	27	41.6	52
Sometimes	30	47.45	28	46.4	58
Often	35	53.64	33	54.4	68
Total	90		88		110

cell j	O_{ij}	E_{ij}	$[O_{ij} - E_{ij}]^2 / E_{ij}$
1,1	25	42.55	7.2386
1,2	27	41.6	5.124
2,1	30	47.45	6.4173
2,2	28	46.4	7.2965
3,1	35	53.64	7.6565
3,2	33	54.4	8.4183

$$\chi^2 = 42.131$$

c)

$$\alpha = 0.01, \quad df = (3-1)(3-1) = 4$$

$$\text{critical value} = 13.277$$

d) since, test statistic > critical value, thus

reject H_0 at $\alpha = 0.05$

Conclusion: There is evidence that age group and online shopping frequency is related.